



Batterjee Medical College
For Science and Technology

Pericardial Diseases

Approach to a Case of Chest Pain

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Learning Objectives

Knowledge & understanding

Describe the clinical features of acute pericarditis, cardiac tamponade (Beck's triad), and constrictive pericarditis.

Describe the management of acute pericarditis, cardiac tamponade, and constrictive pericarditis

Skill

Recognize the clinical signs of cardiac tamponade and constrictive pericarditis on physical examination.

Formulate an appropriate management plan for a patient with pericarditis, cardiac effusion / tamponade and Constrictive Pericarditis.



Acute Pericarditis

Definition, etiology, clinical features, ECG evolution and differentiation from STEMI, laboratory findings, management: NSAIDs/colchicine/glucocorticoids

Pericardial Effusion & Cardiac Tamponade

Definition, etiology, pathophysiology of the rate of fluid accumulation, clinical features including Beck's triad and Ewart sign, diagnostics , ECG/chest x-ray/echocardiography, management and pericardiocentesis

Constrictive Pericarditis

Definition, etiology, clinical features including Kussmaul sign and pericardial knock, diagnostics , cardiac catheterization 'dip and plateau' pattern, management and pericardiectomy

Case Scenario

A 23-year-old man comes to the physician with a 1-week history of sharp, substernal chest pain that is worse with inspiration and relieved with leaning forward . He also had nausea and myalgia. His father has coronary artery disease. His temperature is 37.3°C, pulse is 110/min, and blood pressure is 130/84 mm Hg. Cardiac examination shows a high-pitched rubbing sound between S1 and S2 that is best heard at the left sternal border. ECG shows depressed PR interval and diffuse ST elevations. Which of the following is the most likely cause of this patient's symptoms?

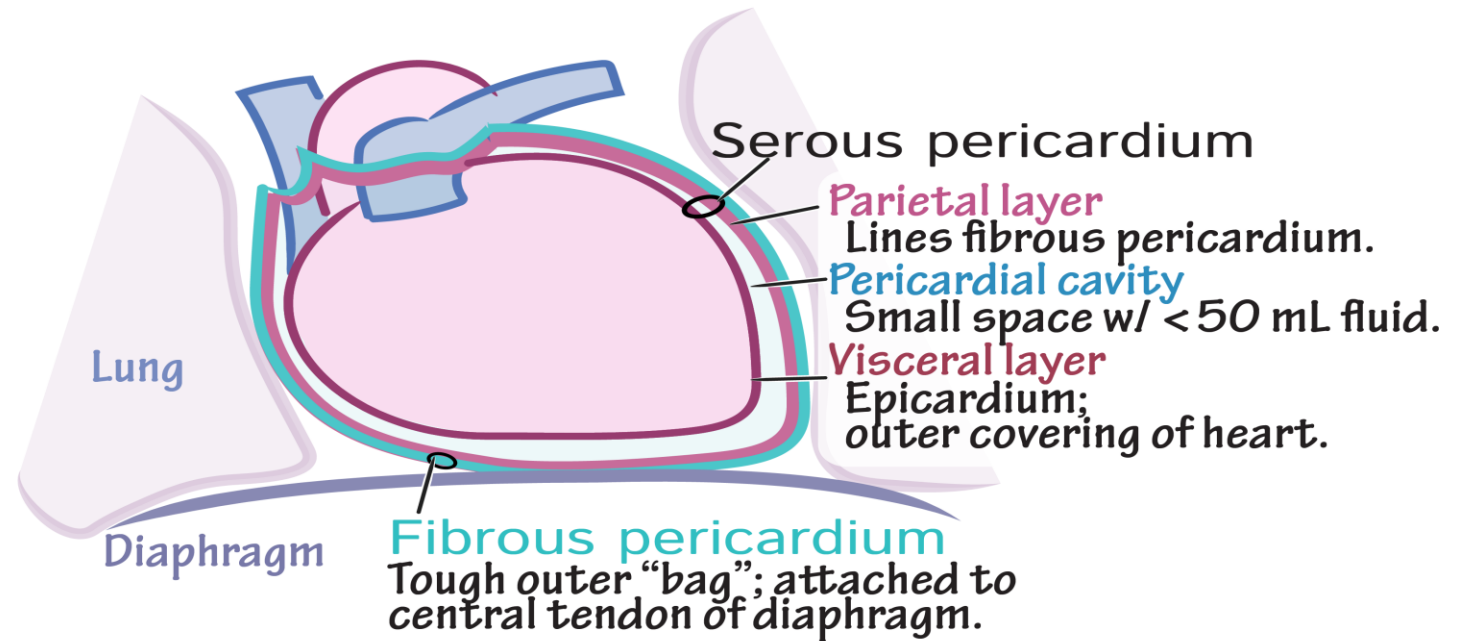
- a. Dressler syndrome
- b. Acute myocardial infarction
- c. Mycobacterium tuberculosis infection
- d. Systemic lupus erythematosus
- e. Acute viral infection

Acute Pericarditis

Definition:

- Acute pericarditis is inflammation of the pericardium
- Exact incidence is unknown because a subclinical course is common

The Pericardium



§ In response to injury, the pericardium increases fluid production; fluid can contain fibrin and inflammatory cells.

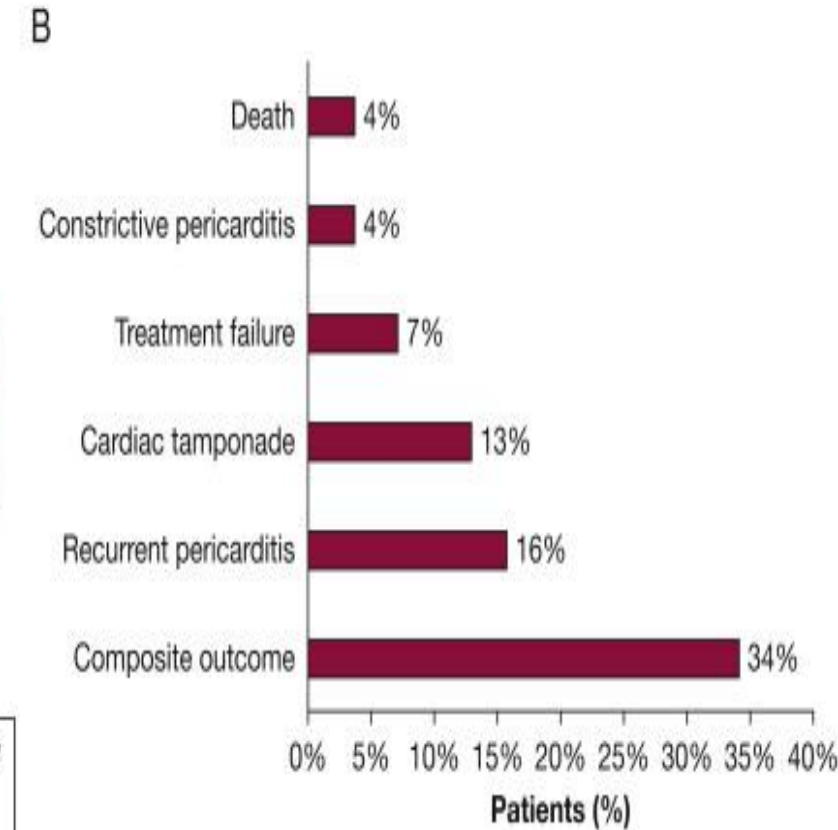
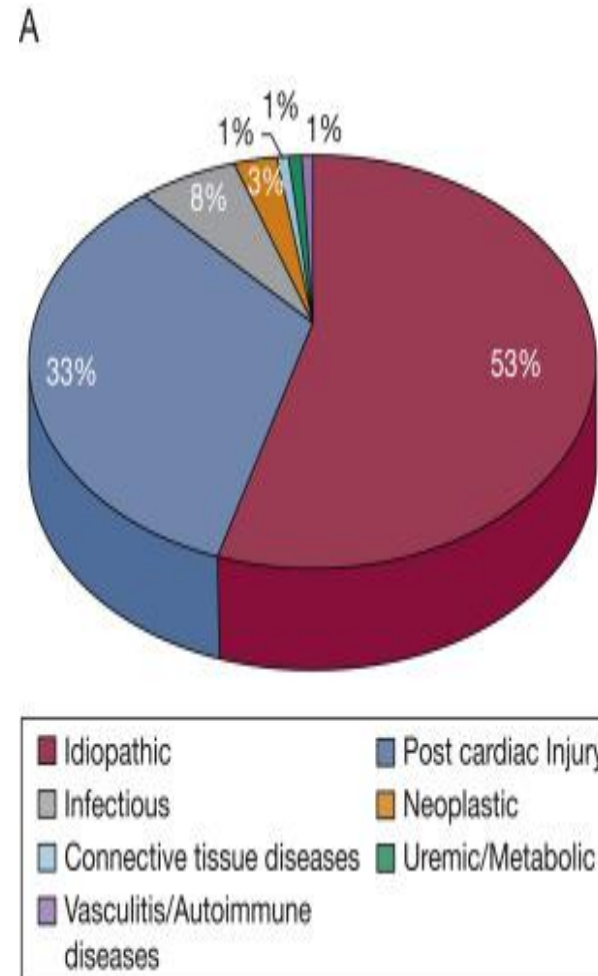


Pericardium can distend to hold this fluid, up to a point.

Acute Pericarditis

Etiology:

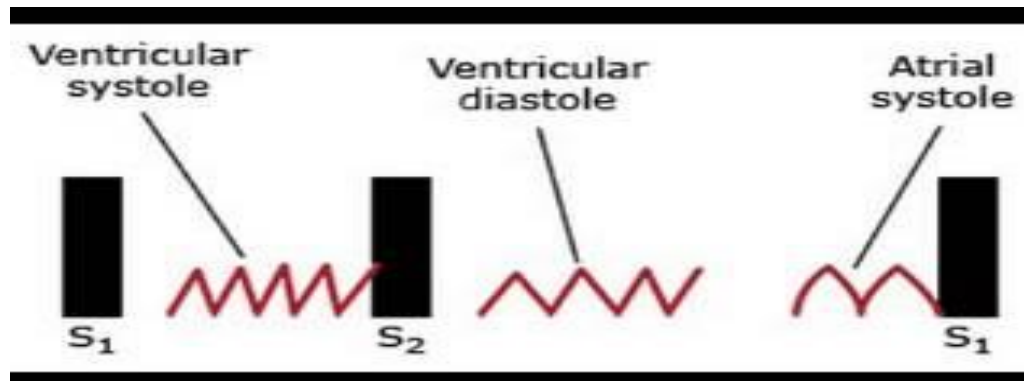
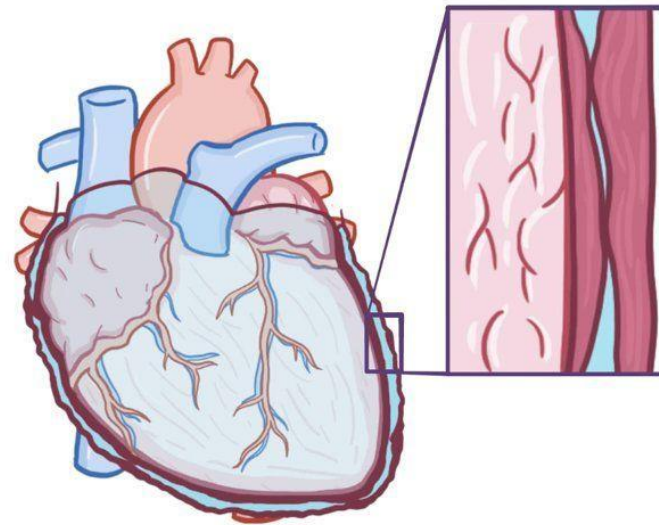
- Idiopathic
- Infectious (**Most commonly viral**)
- Acute MI
- Uremia
- Autoimmune connective tissue diseases
- Neoplasms
- After cardiac surgery
- Radiation
- Trauma



Acute Pericarditis

Clinical Features:

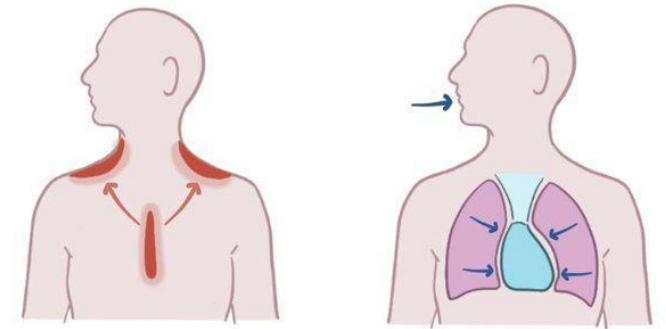
- Positional chest pain
- Pericardial friction rub (3 components)
- Low grade fevers
- Dyspnea, tachypnea, cough
- Hiccups
- Pericardial effusion



SYMPTOMS

* SEVERE, SHARP RETROSTERNAL CHEST PAIN

↳ w/ EACH BREATH



* PAIN WORSENS when SUPINE

↳ IMPROVES when SITTING UPRIGHT & LEANING FORWARD

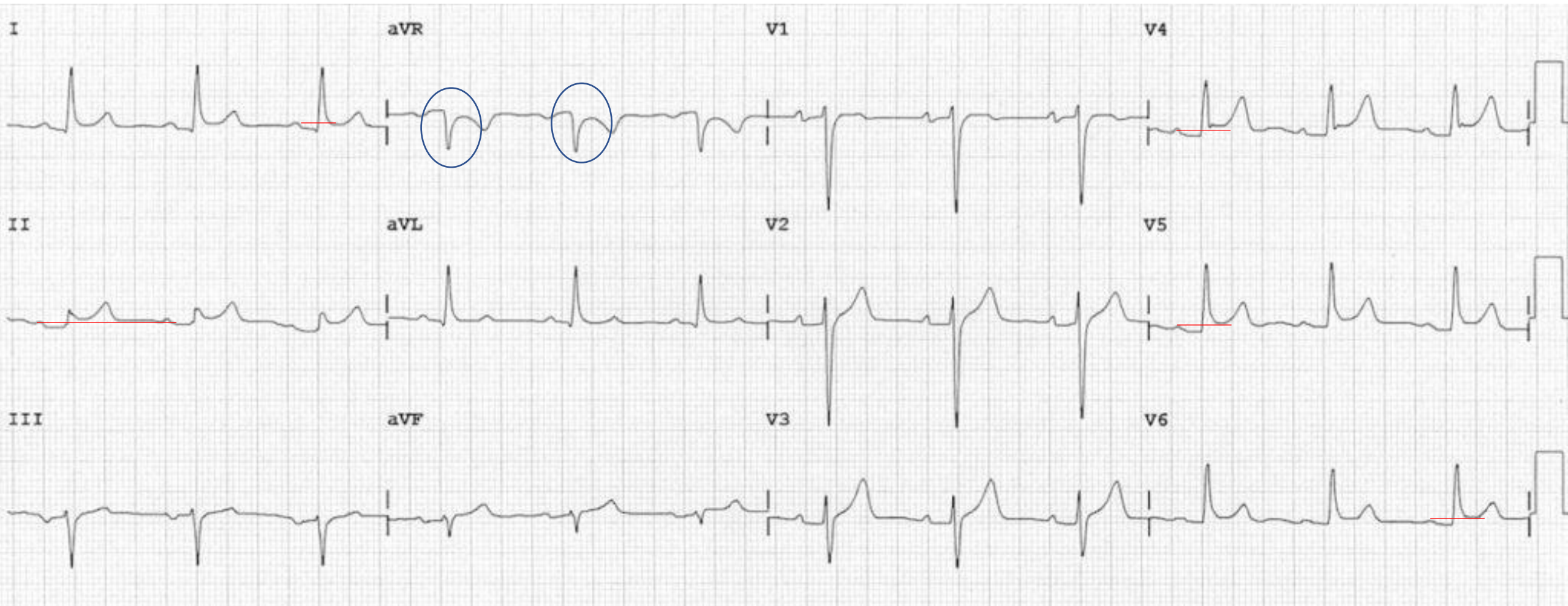
Acute Pericarditis

Diagnostics:

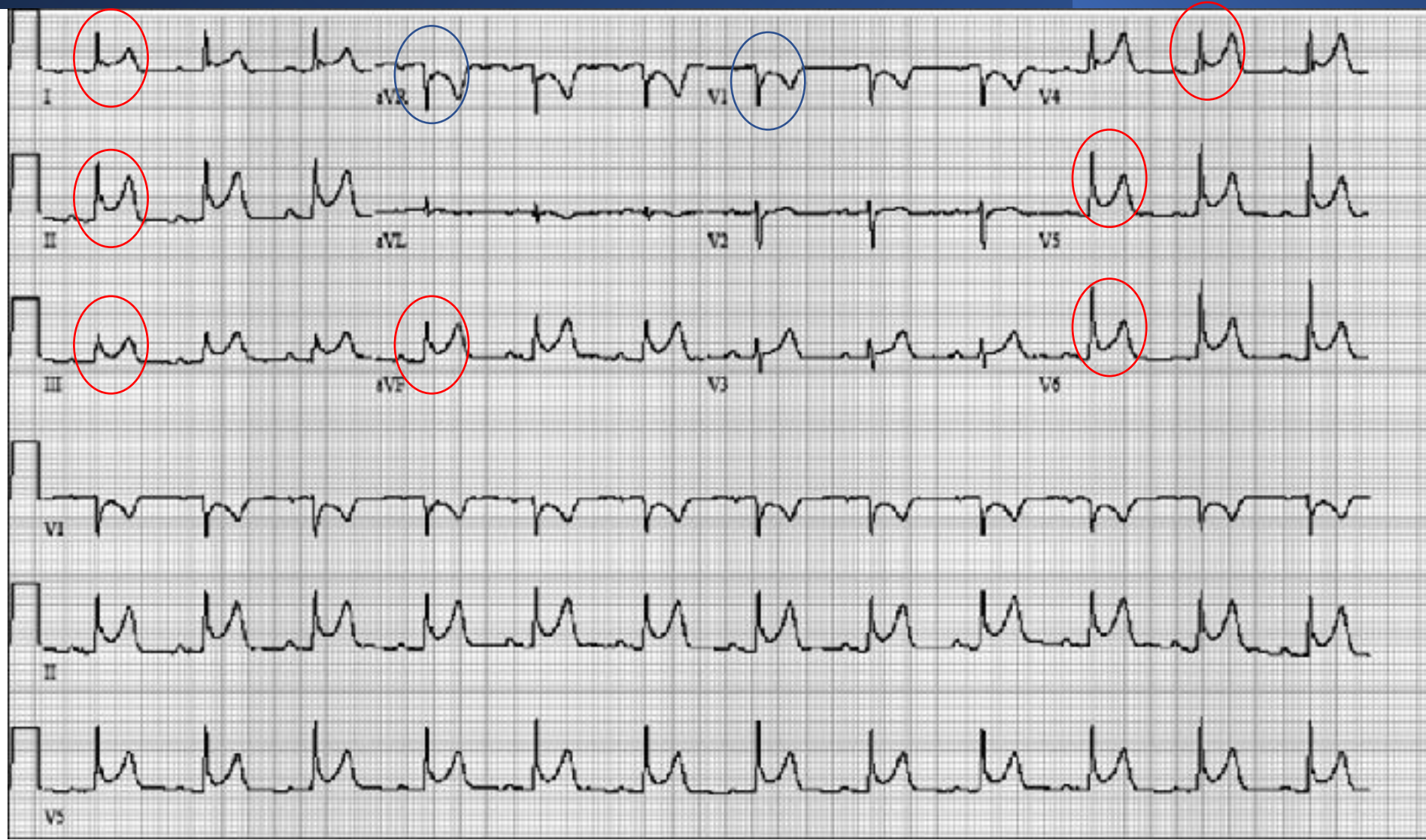
- ECG Changes of acute pericarditis typically evolve over days to weeks
- Widespread ST elevation & PR depression
- Reciprocal ST depression & PR elevation in lead aVR & V1
- Leukocytosis
- ↑ Troponin I (minimal)
- ↑ ESR
- ↑ CRP
- ↑ Creatinine kinase
- Echocardiogram



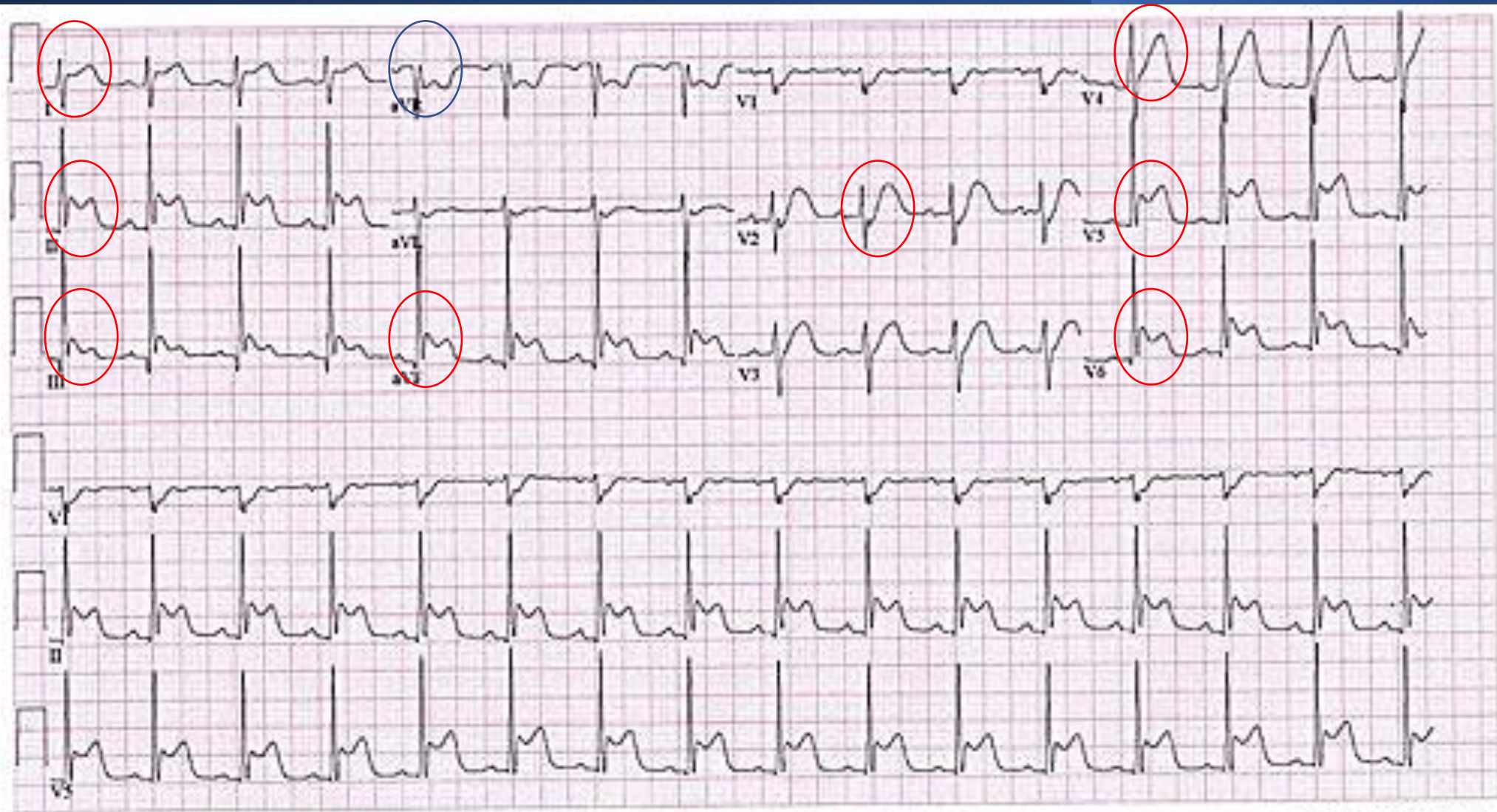
Acute Pericarditis



Acute Pericarditis



Acute Pericarditis



Acute Pericarditis

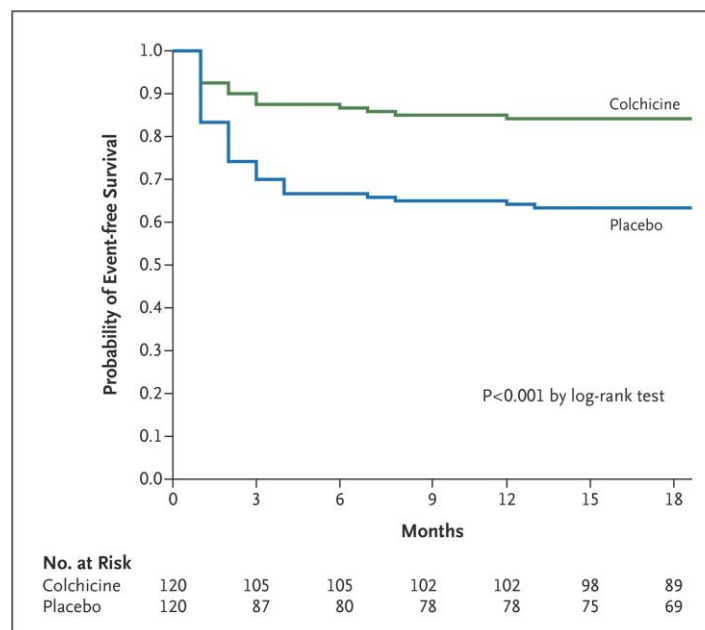
	ST Coving	Which Leads?	ST-T Evolution	PR Segment
Pericarditis		Diffuse	Days-weeks	Depressed
Acute STEMI		Localized	Hours	Normal

Acute Pericarditis

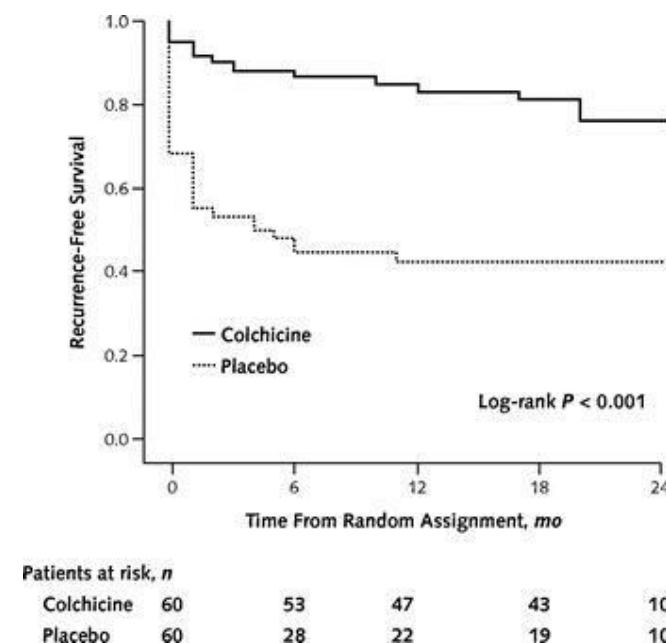
Management:

- First line: High dose NSAID therapy + Colchicine
- Second line: Glucocorticoids
- Gastro-protective therapy

ICAP study



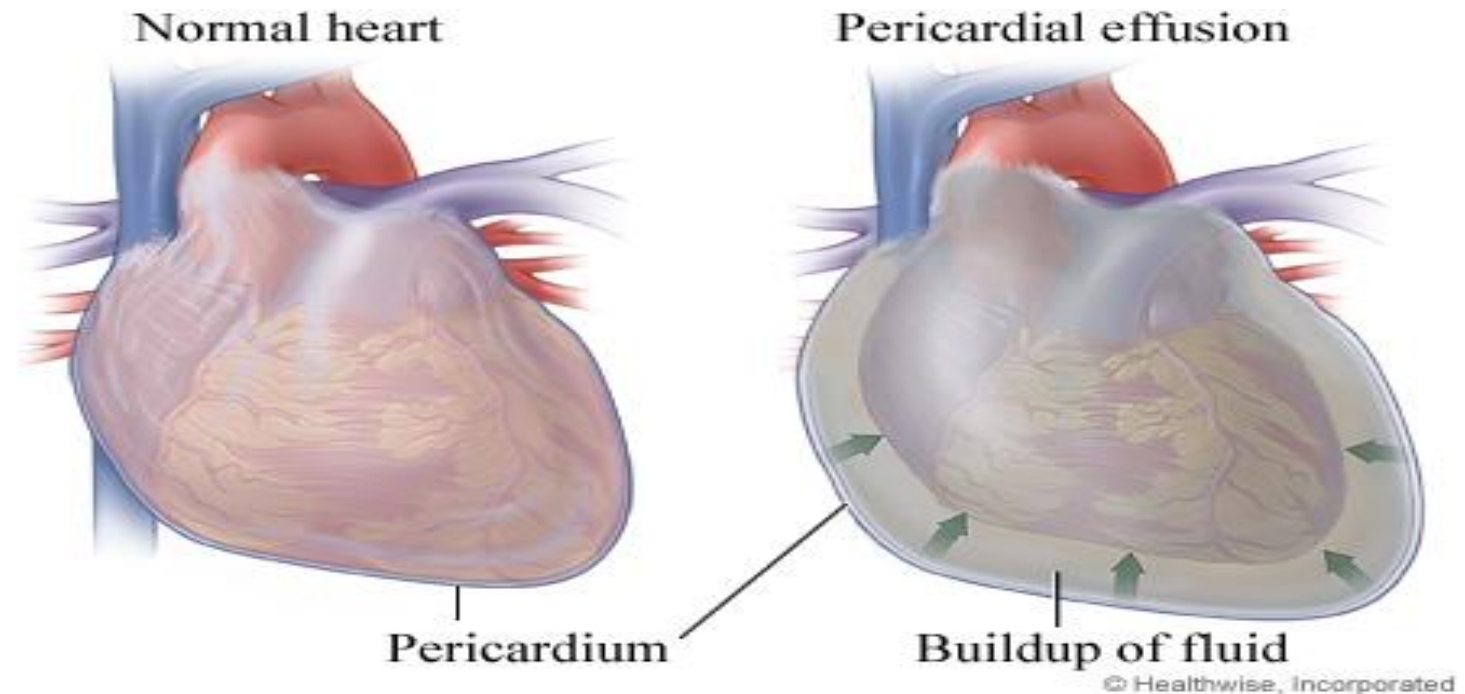
CORP study



Pericardial Effusion & Cardiac Tamponade

Definition:

- **Pericardial effusion**: an abnormal collection of fluid in the pericardial space
- **Cardiac tamponade**: when fluid accumulation results in increased intra-pericardial pressure, leading to cardiac compression, impaired ventricular filling, and reduced cardiac output



Pericardial Effusion & Cardiac Tamponade

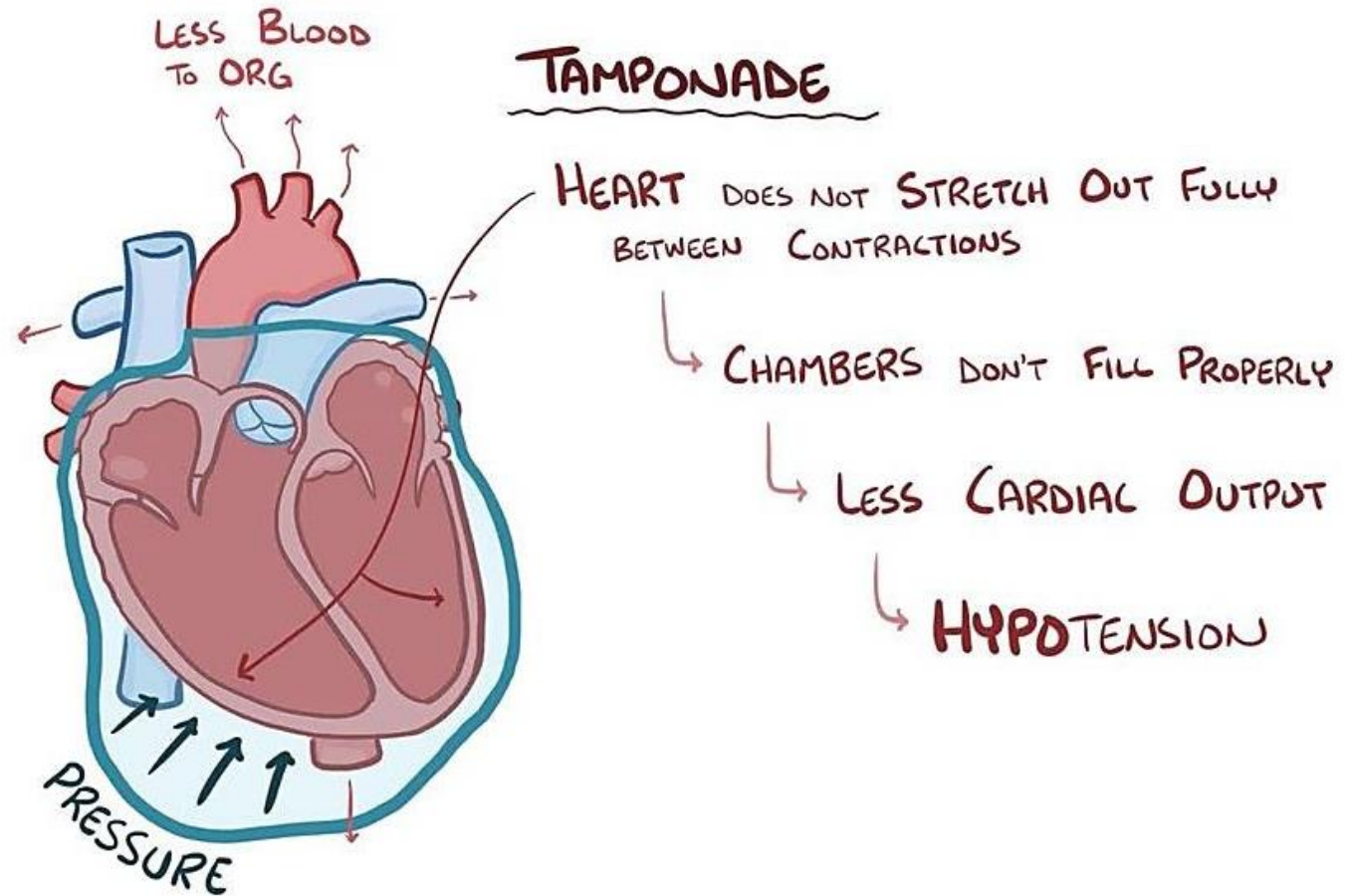
Etiology:

- Cardiac wall rupture
- Chest trauma
- Aortic dissection
- Idiopathic
- Acute pericarditis (Viral, TB, bacterial, etc.)
- Malignancy
- Uremia
- Autoimmune disorders
- Hypothyroidism

Pericardial Effusion & Cardiac Tamponade

Pathology:

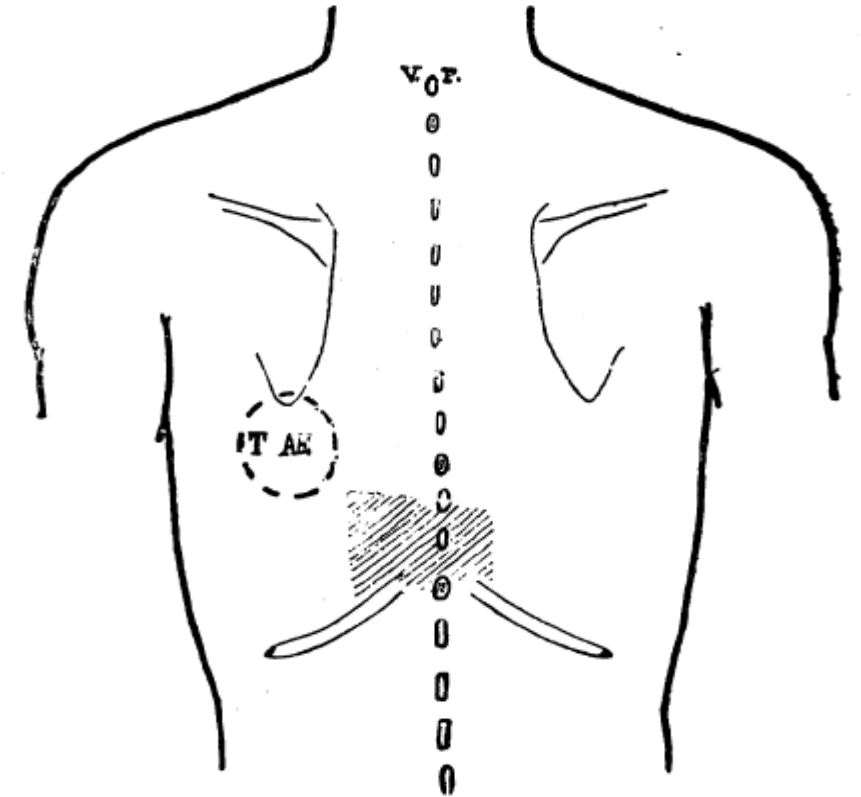
- The hemodynamic consequences of a pericardial effusion depend on the **rate** of accumulation
- Small rapid accumulation can be more dangerous than large slow accumulation



Pericardial Effusion & Cardiac Tamponade

Clinical Features of Pericardial Effusion:

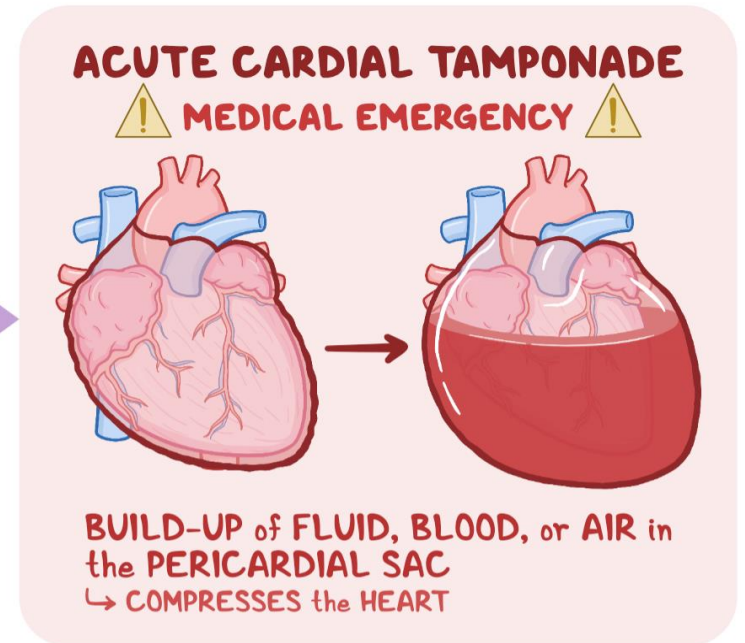
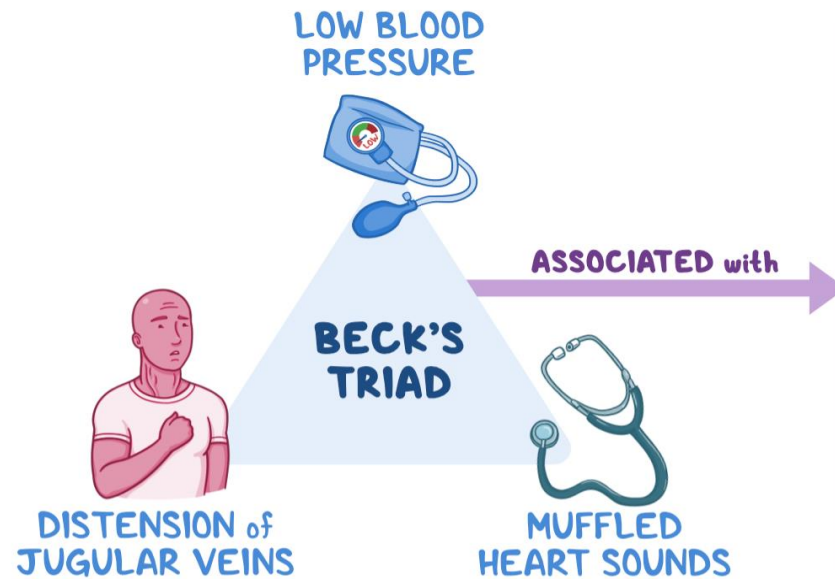
- Dyspnea
- Orthopnea
- Chest pain
- Can cause compressive symptoms
(Hoarseness, cough, Hiccups, etc.)
- Apical impulse is difficult to locate
- **Ewart sign** (Due the compression of lung parenchyma)
 - Dullness to percussion at the base of the left lung with
 - Increased vocal fremitus and
 - Bronchial breathing



Pericardial Effusion & Cardiac Tamponade

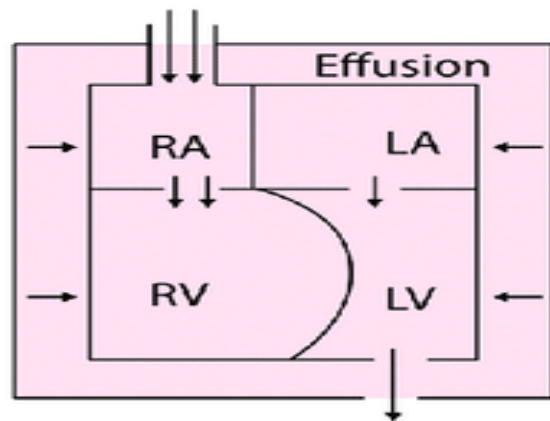
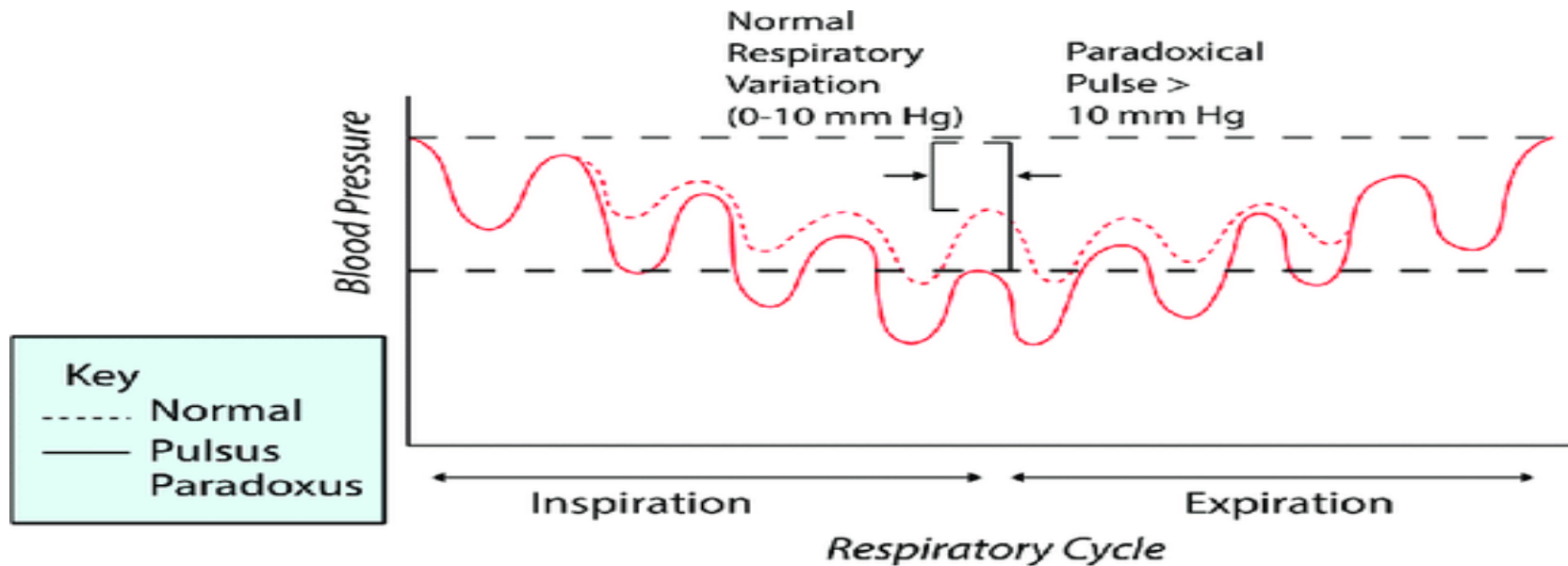
Clinical Features of Pericardial Effusion:

- **Beck's triad**
 - Hypotension
 - Muffled heart sounds
 - Distended neck veins
- Tachycardia
- Tachypnea
- Pulsus paradoxus

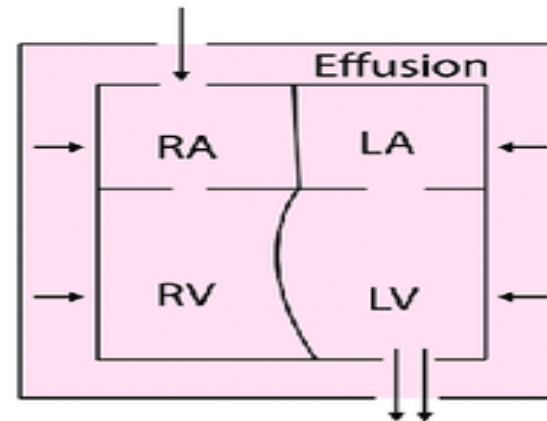


- * NOT EVERYONE with CARDIAC TAMPONADE will show ALL 3 SIGNS of BECK'S TRIAD
- * OTHER SIGNS might appear in CARDIAC TAMPONADE, like TACHYCARDIA and SHORTNESS of BREATH

Pericardial Effusion & Cardiac Tamponade



Inspiration



Expiration

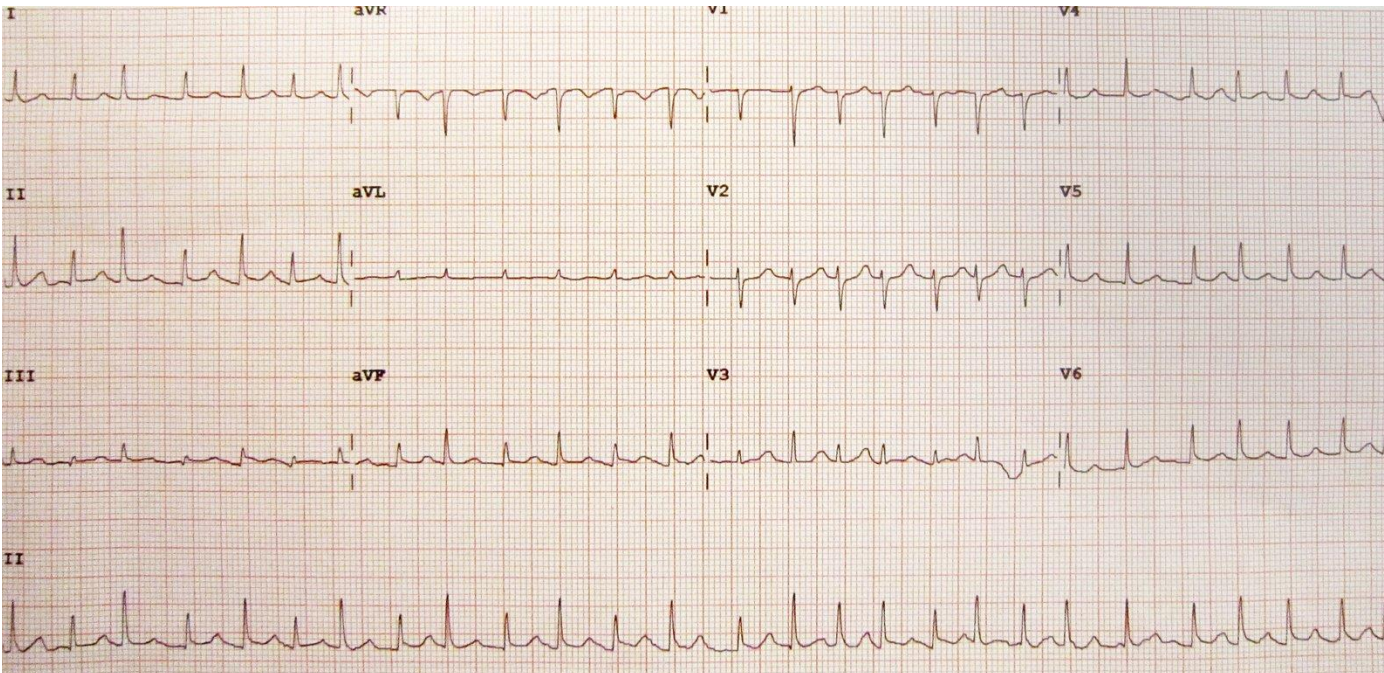
Inspiration
↓
Increased venous return to right heart
↓
Displacement of septum towards left
↓
Decreased filling
↓
Decreased stroke volume

Expiration
↓
Septum moves towards right
↓
Stroke volume increases

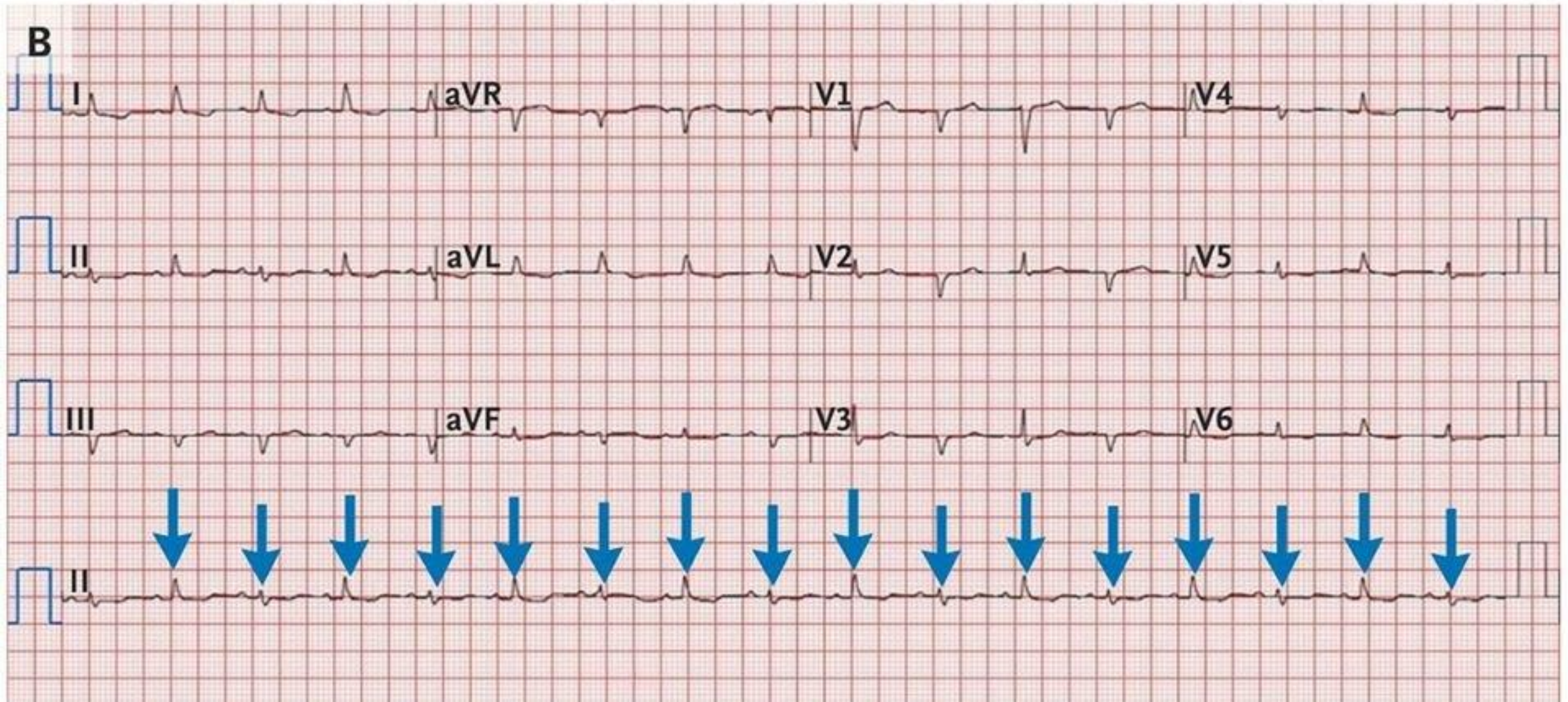
Pericardial Effusion & Cardiac Tamponade

Diagnostics:

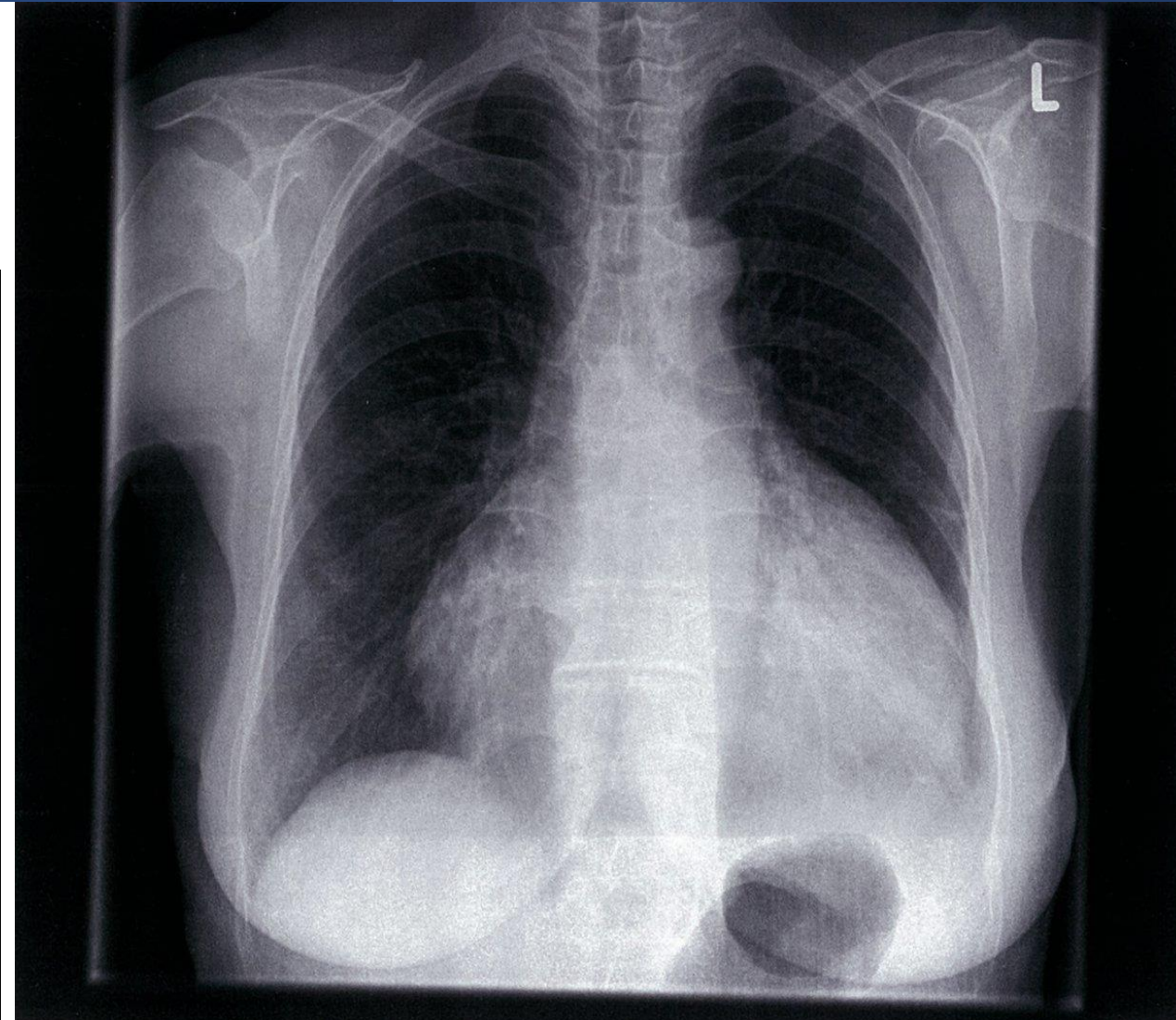
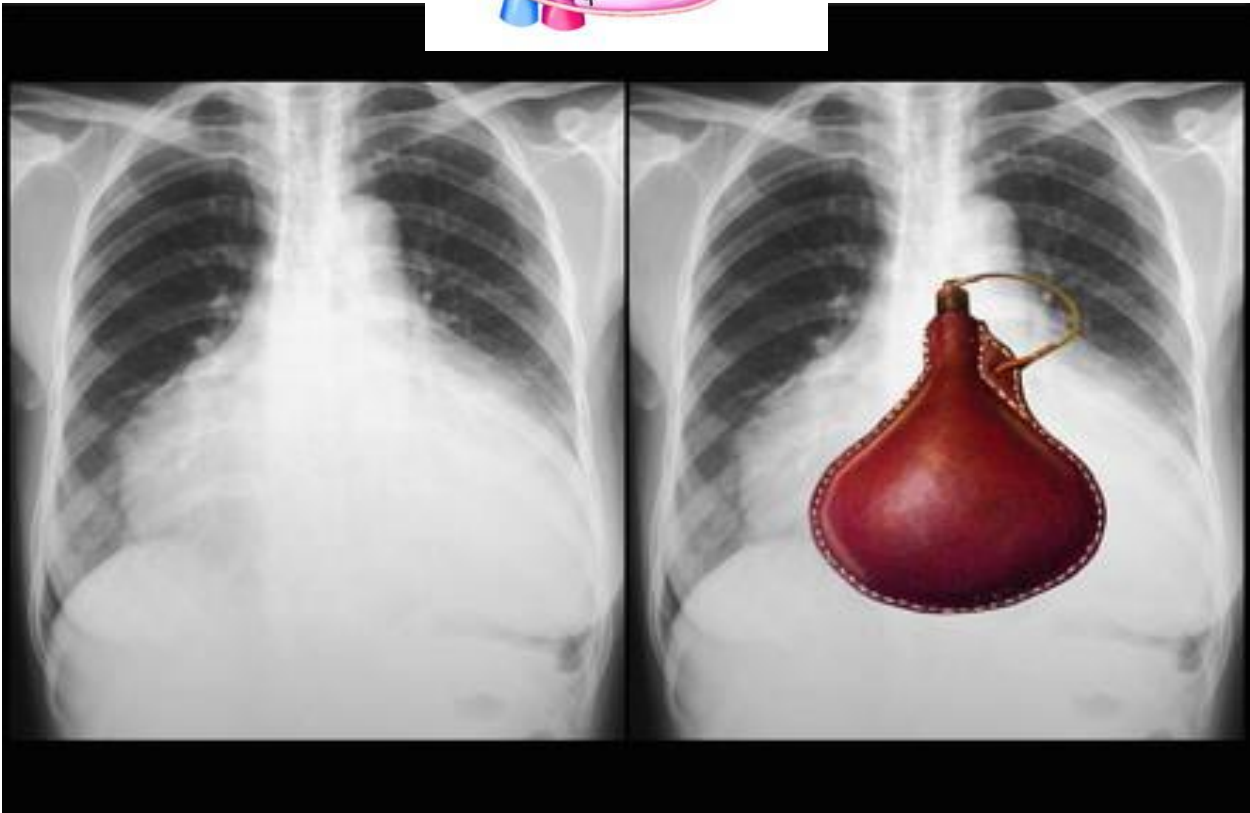
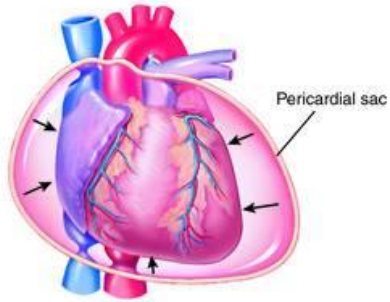
- ECG: Low voltage QRS complexes & electrical alternans
- Chest x-ray: enlarged cardiac silhouette (Water bottle sign)
- Echocardiography (the imaging modality of choice)



Pericardial Effusion & Cardiac Tamponade



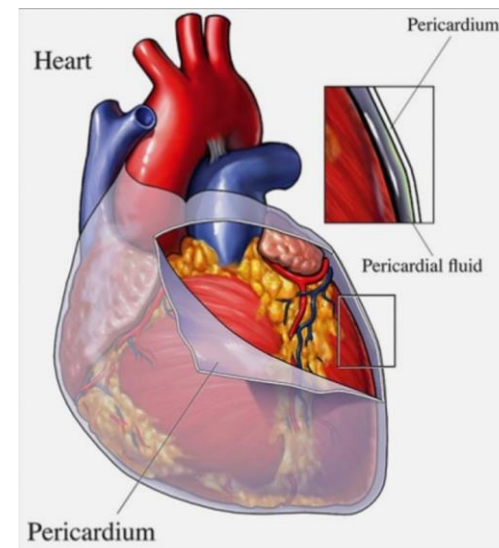
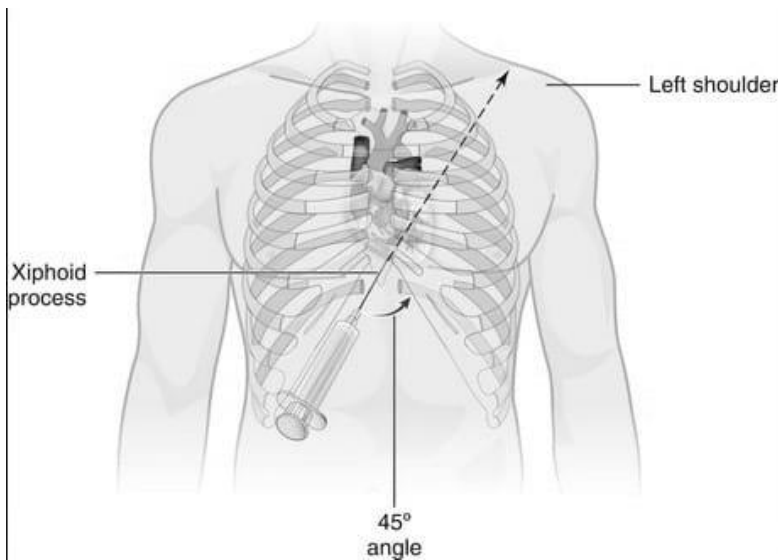
Pericardial Effusion & Cardiac Tamponade



Pericardial Effusion and Cardiac Tamponade

Management:

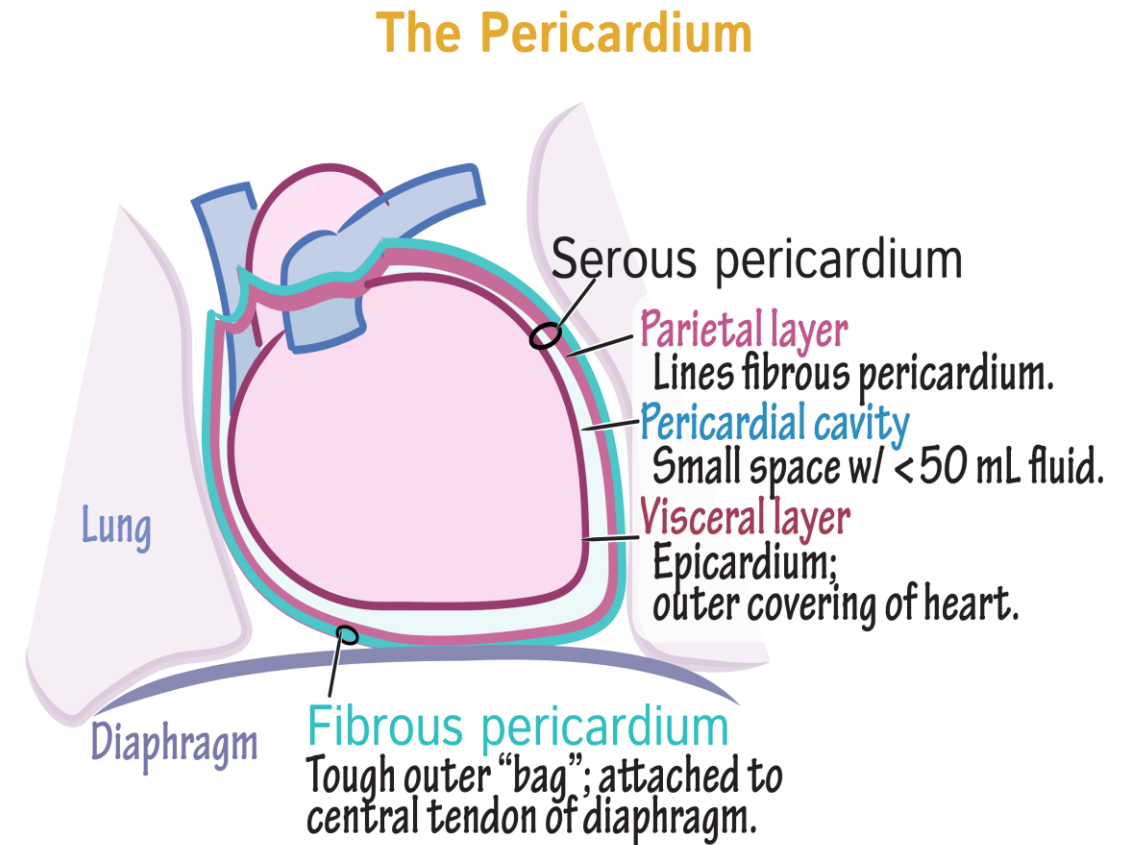
- Routine drainage is unnecessary in the absence of hemodynamic compromise
- Cardiac tamponade is a life-threatening emergency → urgent drainage of the pericardial effusion
- **Fluid resuscitation** should be initiated to increase preload and filling of the cardiac chambers
- For patients with chronic, recurring effusions → surgical creation of a pleuropericardial window



Constrictive Pericarditis

Definition:

- Pericardial constriction caused by pericardial inflammation (scarring, fibrosis, & adhesion of pericardium layers)
- Characterized by a rigid, scarred pericardium that limits diastolic filling of the ventricles, resulting in increased intra-cardiac pressures



Constrictive Pericarditis

Etiology:

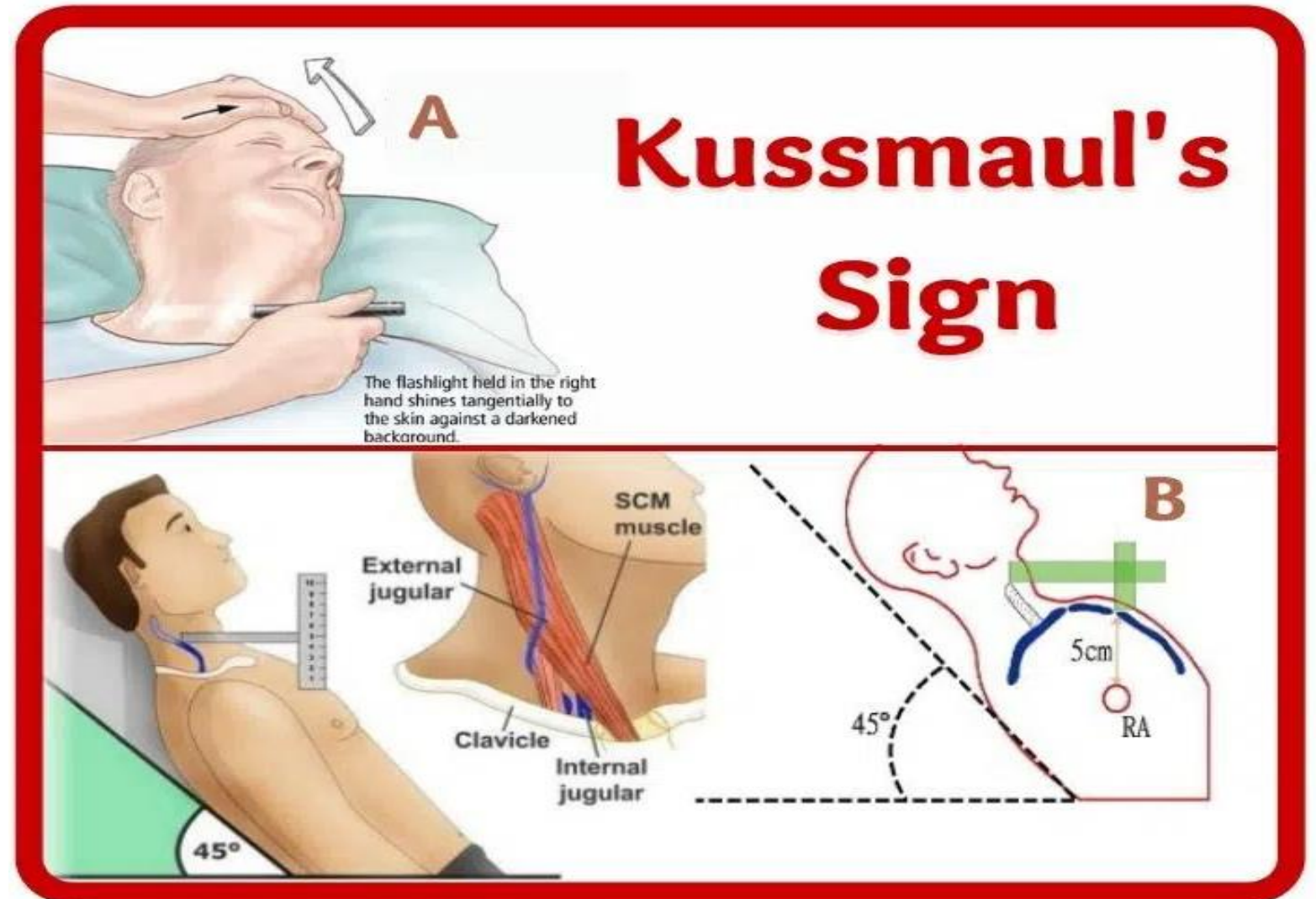
- Infections
- Prior cardiac surgery
- Trauma
- Irradiation
- Connective tissue disorders
- Uremia
- Neoplastic involvement of the pericardium
- Tuberculous pericarditis

Constrictive Pericarditis

Clinical Features:

- Fatigue
- Dyspnea on exertion
- LE edema
- Tachycardia
- Jugular vein distention
- Kussmaul sign
- Ascites
- Pericardial knock

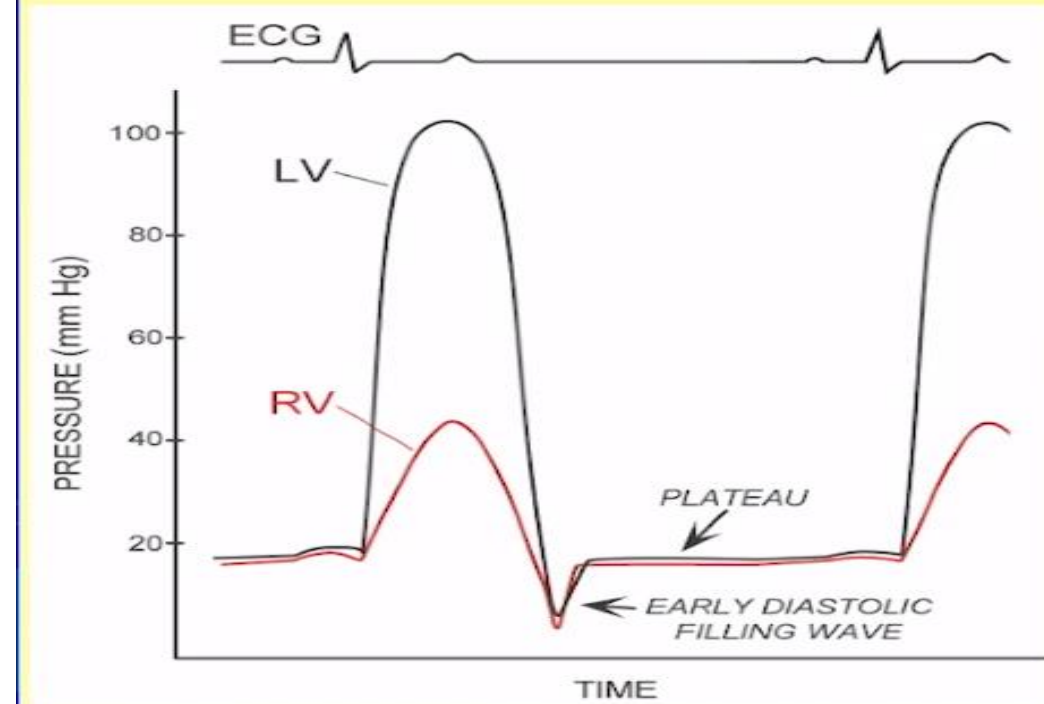
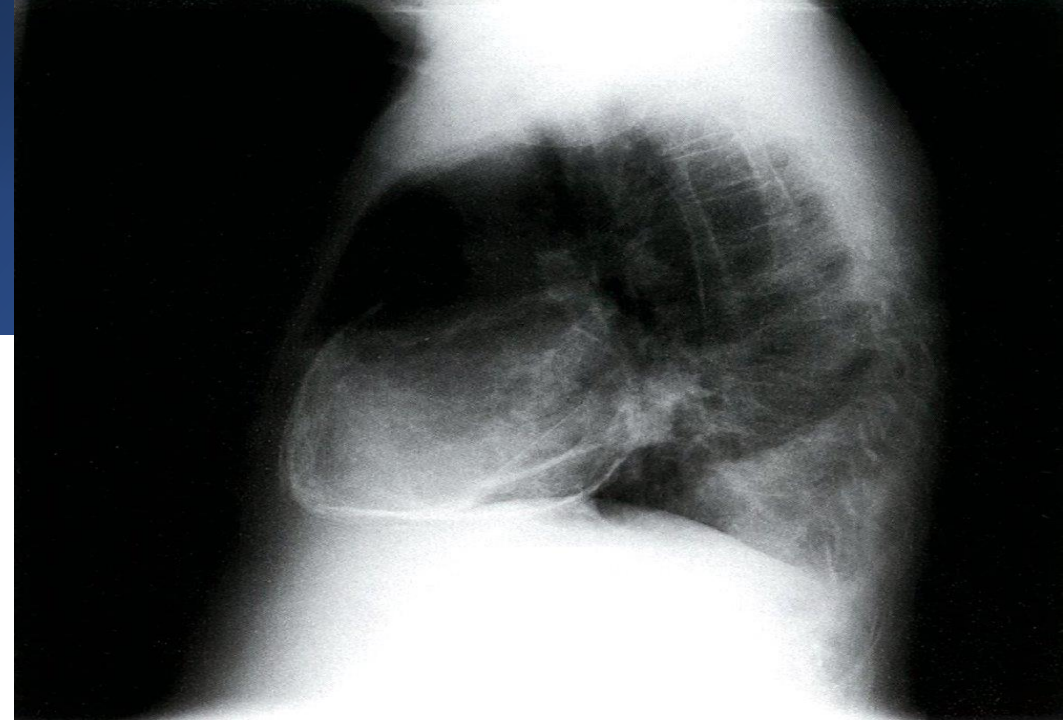
https://youtu.be/VCQKA_SPCGs?t=20



Constrictive Pericarditis

Diagnostics:

- ECG: low QRS voltage, left atrial enlargement
- Chest x-ray: Pericardial calcifications
- Echocardiogram: Pericardial thickness
- Cardiac CT/MRI
- Cardiac Catheterization:
 - Elevation of diastolic pressure in ventricles
 - Equalization of diastolic pressure in ventricles
 - “Dip and plateau” pressure pattern (square root sign)

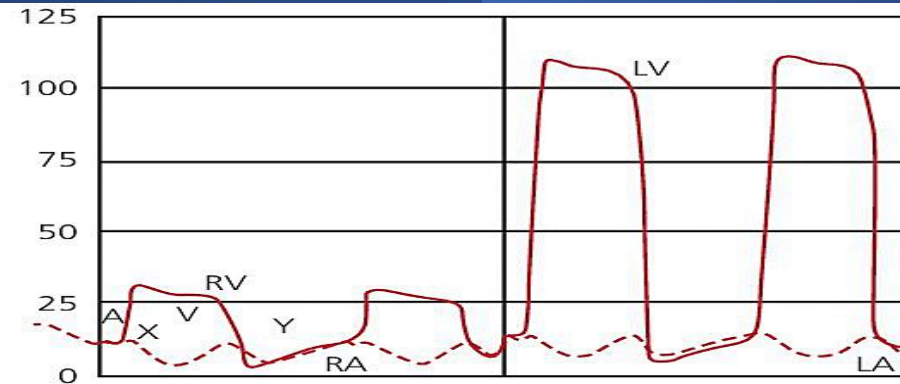
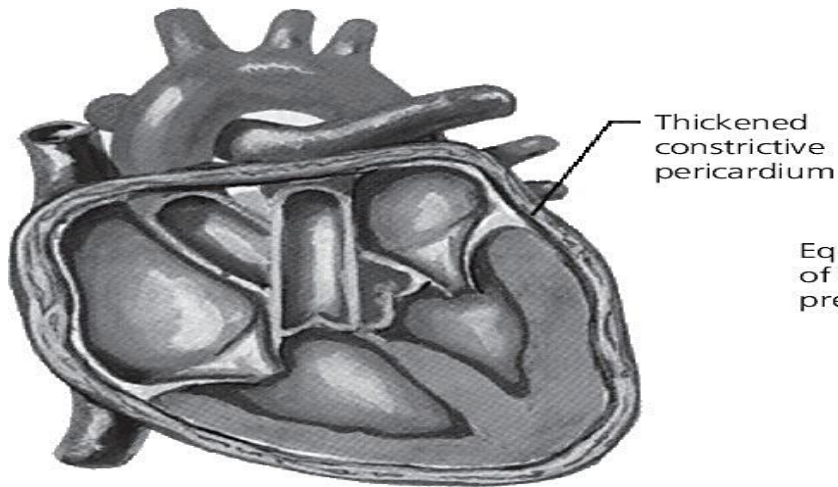


Constrictive Pericarditis

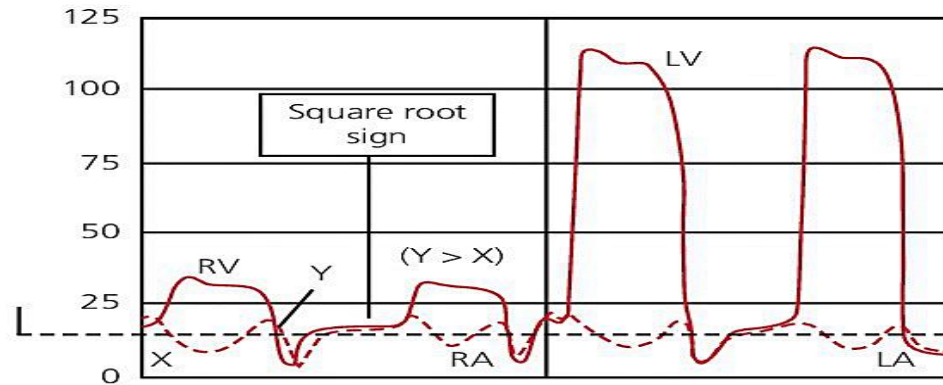
Normal



Constrictive pericarditis



Atrial contraction reduces atrial volume and increases atrial pressure (A wave). Ventricular contraction causes initial small C wave and as atrial relaxation ensues, atrial enlargement occurs with pressure decrease (X descent). Passive atrial filling causes V wave until AV valves open and pressure rapidly drops (Y descent) as ventricles relax. Following ventricular systole, an active and passive filling phase follows—pressure lowest in active phase.



High atrial pressure when the AV valves open results in rapid early filling (rapid Y descent) until filling abruptly stops (square root sign). There is equalization of late diastolic pressures. The RV diastolic is usually $> \frac{1}{3}$ RV systolic pressure.

Constrictive Pericarditis

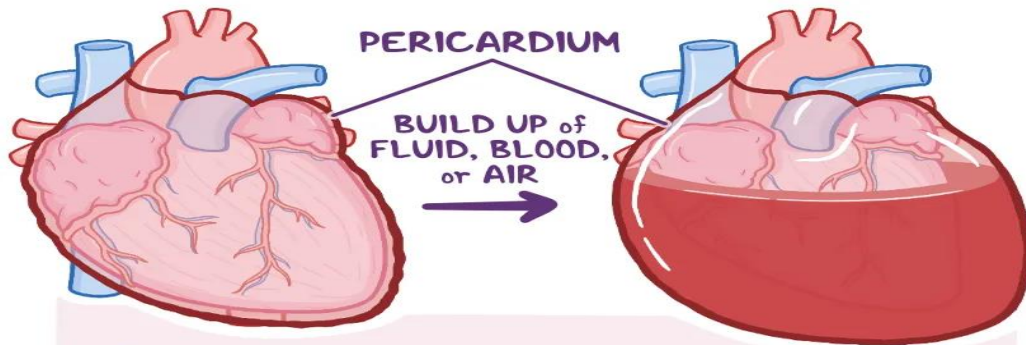
Management:

- **Pericardiectomy** is the only definitive treatment for constrictive pericarditis
- Na restriction and diuretics for those who are not surgical candidates

To Summarize

ACUTE PERICARDITIS * INFLAMMATION of the PERICARDIUM 	DIAGNOSIS ↑ WBC, ESR, & CRP ECG - ST SEGMENT ELEVATION in SEVERAL LEADS - PR SEGMENT DEPRESSION	TREATMENT ANTI-INFLAMMATORY MEDICATIONS - can COMBINE w/ COLCHICINE
PERICARDIAL EFFUSION * 200 mL of FLUID ACUTELY * > 1L if SLOWLY 	SMALL → ASYMPTOMATIC - UNLESS ASSOCIATED w/ PERICARDITIS LARGE → CARDIAC TAMPONADE	SMALL → NO TREATMENT HOSPITAL ADMISSION PERICARDIOCENTESIS VOLUME EXPANSION
CONSTRUCTIVE PERICARDITIS * FIBRINOUS SCARRING & CALCIFICATION of the PERICARDIUM 		PERICARDECTOMY

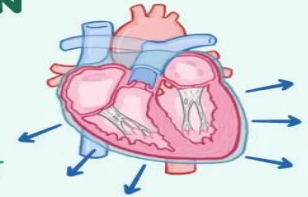
To Summarize



⚠️ **CONSIDERED a MEDICAL EMERGENCY**
CAN PROGRESS to CIRCULATORY SHOCK and CARDIAC ARREST

PERICARDIAL EFFUSION

- PERICARDIAL FLUID BUILDS up SLOWLY OVER TIME
 - ↳ ALLOWS PERICARDIUM to STRETCH OUT to ACCOMMODATE BIGGER VOLUMES of FLUID w/o COMPRESSING HEART



SYMPTOMS



CHEST PAIN



SHORTNESS of BREATH



COMPRESSION of NEAR STRUCTURES



BECK'S TRIAD



DISTENSION of JUGULAR VEINS



MUFFLED HEART SOUNDS

ACUTE PERICARDIAL TAMPONADE

- SUDDEN FLUID ACCUMULATION
- PERICARDIUM CANNOT ADJUST
 - ↳ DRAMATIC INCREASE in PRESSURE INSIDE PERICARDIAL SAC

CAUSES



CHEST TRAUMA



RUPTURED AORTA



RUPTURED of VENTRICLE AFTER a HEART ATTACK

References & further reading

- Step-up to medicine 5th edition
- Harrison's Principles of Internal Medicine 12th edition
- Amboss
- Osmosis.org

Thank You